

Part 1: ACME Corp Enterprise Context Document

DOCUMENT TITLE: ACME Corp - AI Architecture, Standards, and FinOps Policies

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OWNERS: Enterprise Architecture Board & Cloud FinOps

1. Organization Overview

ACME Corp is a leading manufacturing and media conglomerate. Our Google Cloud footprint relies heavily on advanced machine learning for manufacturing quality assurance, massive media processing, and enterprise search.

2. Key AI & Compute Workloads

- **Manufacturing QA Engine (Tier-1):**
 - **Description:** Real-time defect detection on assembly lines.
 - **Architecture:** Vertex AI AutoML Image Classification.
 - **Requirements:** Models must remain deployed 24/7, even during seasonal assembly line maintenance, to support zero-latency unpredictable test runs. Cold-starts are strictly prohibited.
- **Media Recommendation Engine (Tier-1):**
 - **Description:** Suggests media to users based on historic viewing data.
 - **Architecture:** Vertex AI Vector Search (e2-standard-16) and Vertex AI Feature Store.
- **Enterprise Knowledge Graph (Tier-2):**
 - **Description:** Internal search engine for corporate and media documents.
 - **Architecture:** AgentSpace Enterprise Plus.
- **Legacy 3D Rendering (Tier-3):**
 - **Description:** Overnight batch rendering of historic media assets.
 - **Architecture:** Raw Compute Engine VMs with attached GPUs.

3. Cloud Infrastructure & GPU Standards

- **GPU Standardization:**
 - All **modern AI workloads** (inference and training via Vertex AI) must use Nvidia L4 GPUs, standardized in the us-central1 (Iowa) and us-east1 (South Carolina) regions.
 - **Legacy media processing** workloads are restricted to Compute Engine Nvidia Tesla V100 GPUs due to hardcoded legacy CUDA library dependencies. Do not attempt to migrate these to L4s without a full application rewrite.
- **Compute Provisioning:** * Spot VMs (Preemptible instances) are strictly prohibited for the overnight Legacy 3D Rendering batches, as interruptions corrupt the output files.
- **Vector Search & Feature Store:**
 - Vector Search indexes must be served on memory-optimized or high-memory machines (e.g., e2-standard-16) to ensure low-latency recall for the Media

Recommendation Engine.

4. Software Licensing Policies (SaaS / Agentspace)

- **Full-Time Employees (FTEs):** Must be provisioned on 1-year or 3-year term commitments for Agentspace Enterprise Plus and NotebookLM Enterprise to maximize long-term savings.
- **Contractors / Temporary Staff:** Must strictly be provisioned on 1-month term SKUs. We regularly surge temporary staff for 30-90 day data review projects; committing to 1-year terms for these seats results in abandoned licenses and wasted capital.